

(3 Hours)

Total Marks: 80

NB:

- 1) Question number 1 is compulsory
- 2) Attempt any three out of the remaining five questions.
- 3) Assume suitable data if necessary and justify the assumptions.
- 4) Figures to the right indicate full marks

Q1

A What is the difference between data science and data analytics?

10

B What is ANOVA? Brief about benefits of ANOVA technique.

10

Q2

A What are Type I and Type -II errors? Give examples.

10

B Explain the data science tasks with proper examples.

10

Q3

A Describe the terms: Cross Validation, K-fold cross validation, leave-1 out and Bootstrapping.

10

B Calculate the coefficient of correlation for the following data with Karl Pearson's method.

10

X	15	18	20	28	34
Y	40	42	46	50	52

Q4

A Find Bowley's coefficient of skewness of the following series.

10

Profit (in crores)	4-8	8-12	12-16	16-20	20-24
No. of films	4	10	15	8	3

B What are the pros and cons of an Auto Regressive Integrated Moving Average (ARIMA) model? Explain with proper examples.

10

Q5

A Explain the steps to build a product recommendation model in detail.

10

B What is Hypothesis testing? Write about the different types of Hypothesis testing.

10

Q6

Write a note on any FOUR:

20

A. Data Visualization

B. Applications of Data Science

C. Data Exploration

D. Taxonomy of time series forecasting methods

E. Outlier detection methods